

Abdoali Mazhar Zakir

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PROFESSIONAL SUMMARY

Final-year Software Engineering (Honours) student at UNSW who specialises in building scalable data systems and applied machine learning solutions. Shipped production-ready applications across the stack, including financial knowledge graphs and AI-driven platforms. Streamlined deployment pipelines to cut manual workflows by 30% and improved system reliability. Currently working on developing machine learning models for complex Citation Link prediction.

EDUCATION

Bachelor of Engineering (Honours) in Software Engineering
University of New South Wales

September 2024 - December 2026

Bachelor of Science in Computer Science & Engineering
Manipal Academy of Higher Education, India

July 2022 - May 2024

EXPERIENCE.

Data Analyst Intern

Futures Sport & Entertainment

July 2026 - September 2026

- Joined the Data Analytics team at Futures Sport & Entertainment, supporting data-driven sponsorship and audience insight work across sport and entertainment clients.
- Working across data collection, cleaning, and analysis to support client-facing reporting and insight generation, three days per week through the internship period.

Data Engineer Intern

Tata Consultancy Services (TCS)

January 2026 - February 2026

- Engineered a Financial Crime Compliance knowledge graph by transforming large-scale transaction and account datasets into structured graph entities, delivering a scalable framework for deeper analysis of complex financial networks.
- Applied graph similarity algorithms to detect hidden relationships between accounts, strengthening fraud-risk detection capability by surfacing previously unidentified connections across financial networks.
- Presented technical data findings and graph-based methodology to industry stakeholders, translating complex metrics into plain-language risk insights that improved stakeholder decision-making.
- Collaborated within a multi-disciplinary Agile engineering team on project planning and milestone reviews, contributing to the successful delivery of final technical presentations to industry stakeholders.

PROJECTS

Sentinel ETL Self-Healing Data Platform
Personal Project

June 2026

- Architected an enterprise-inspired ETL self-healing data quality engine (Python/FastAPI) that automatically detects and remediates data inconsistencies, with full audit logging and data lineage tracking for every transformation.
- Engineered a platform implementing Medallion Architecture (Bronze/Silver/Gold layers) to ingest, validate, repair, and transform datasets automatically.
- Developed a Next.js/TypeScript/Tailwind analytics dashboard to visualise data quality scores, pipeline health, and dataset history. Used DuckDB, Pandas, and PyArrow for the data processing layer, PostgreSQL for metadata storage. [GitHub: [Sentinel-ETL](#)]

- Automated data ingestion and cleansing workflows on AWS Lambda, unifying disparate weather variables rainfall, frost, temperature, and wind into a standardised pipeline for downstream forecasting models.
- Architected responsive frontend features using Next.js (App Router), including dashboards and recommendation views, with state management and API integration.
- Implemented server-side generative AI features using OpenAI API (gpt-4o-mini), including a viticulture assistant and forecast-based recommendation system using structured outputs.
- Collaborated in an Agile team, contributing to API design, system integration, and technical documentation, including testing strategy and deployment planning.

EXTRACURRICULAR ACTIVITIES

Software Developer Enigma Technical Society	September 2023 – May 2024
• Built internal reporting and automation scripts in Python to track and analyse society operations data, cutting manual reporting effort by 30%. • Designed and maintained a PostgreSQL data layer for event and membership data, ensuring clean, query-ready datasets for a 6-member Agile team. • Automated data-refresh and deployment workflows using GitHub Actions CI/CD pipelines, reducing manual errors in recurring data updates. • Contributed to brainstorming sessions and technical documentation, reinforcing code quality and collaborative development practices.	

CORE COMPETENCIES

Languages: Python, JavaScript, C++, SQL, Cypher, HTML/CSS.

Frameworks & Libraries: Node.js, React, Jest, Pytest, PyTorch

Databases: PostgreSQL, MySQL, Neo4j(graph)

Tools & Platforms: Docker, AWS (Terraform, S3), Databricks, Postman, Neo4j

VOLUNTEERING AND SOCIETIES

Lead Volunteer Voluntary Service Overseas (VSO), Manipal	June 2023- May 2024
Secretary Social Football Society, UNSW	March 2024- Present
Sports Captain International Centre for Applied Science, Manipal University	June 2023- May 2024